Chemical Engineering Process Calculator

C Programming Based Engineering Toolkit

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Introduction

Engineering calculation are essential in chemical industries.

This project develops a simple software tool using C programing to perform basic chemical engineering calculations.

Objective

Apply programing in chemical engineering.

Automate engineering calculations.

Improve problem-solving skills.

Understand industrial parameter calculations.

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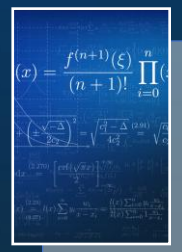
Technologies Used

C Programing

Engineering Concepts



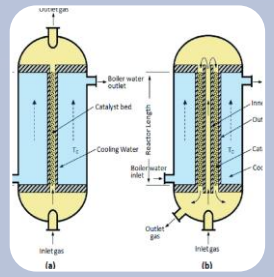
Mathematical Modeling

Features of Project

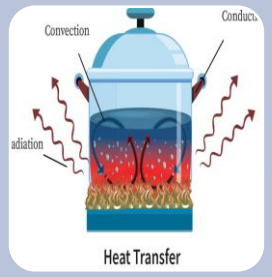
Temperature Converter

Pressure Converter

Flow Rate Calculator



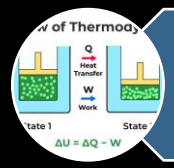
Reactor Conversion Calculator



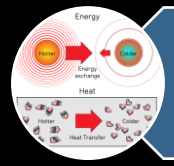
Heat Transfer Calculator

Temperature Module Explanation

Application



Thermodynamics



Heat transfer

Conversion Formulas:

- $K = C + 273.15$
- $F = (C * 1.8) + 32$

```
// Example C Implementation
float celsius_to_kelvin(float C) {
    return C + 273.15;
}
float convert_to_ferenheit(float C) {
    return (C * 1.8) + 32;
}
```

Flow & Reactor Module

Mass Flow:

$M = \rho \times Q$

M = Mass(Kg)

$\rho = Density(Kg/m^3)$

Q = Vol. Flow Rate(m^3/s)

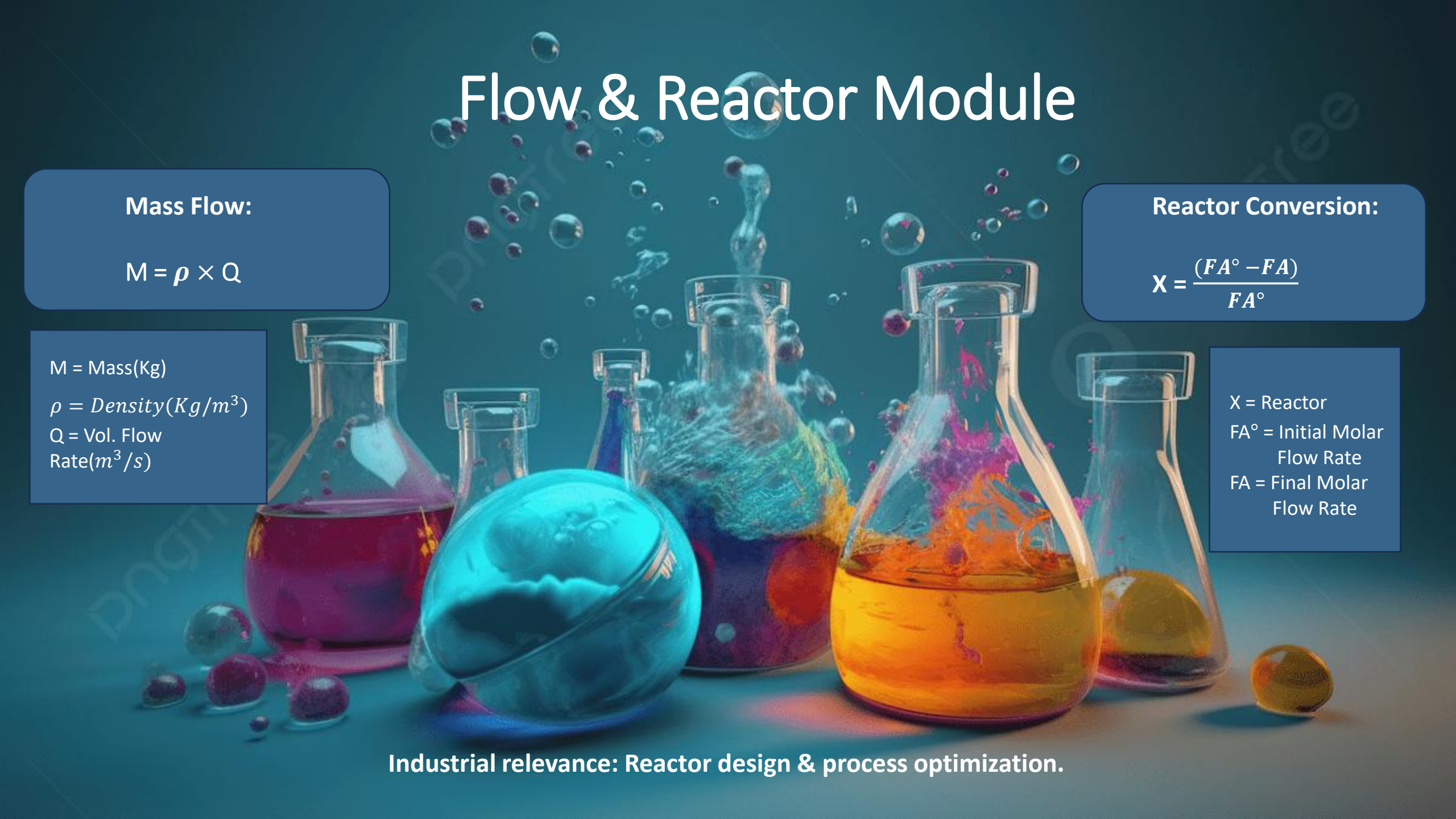
Reactor Conversion:

$$X = \frac{(FA^\circ - FA)}{FA^\circ}$$

X = Reactor

FA° = Initial Molar Flow Rate

FA = Final Molar Flow Rate



Industrial relevance: Reactor design & process optimization.

Heat Transfer Module

Heat Transfer:

Q = m * cp *ΔT

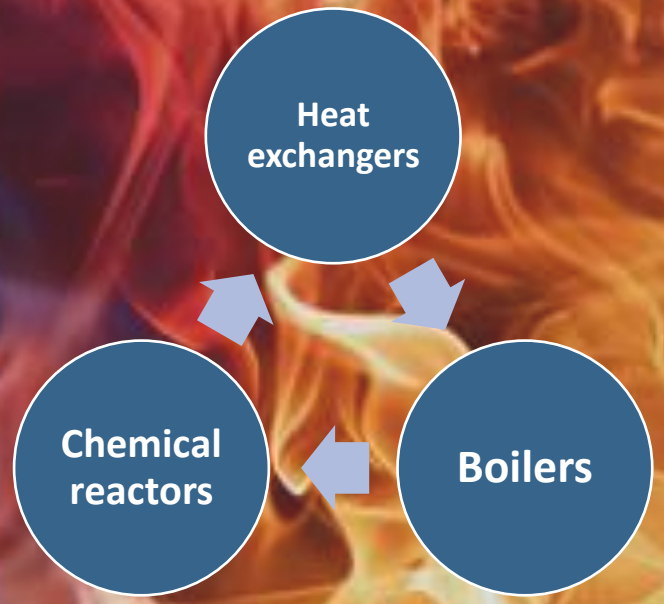
Q = Heat Transferred(J)

m = Mass(Kg)

c_p = Specific (Heat(J /Kg.K)

$\Delta T \quad = \quad T\ empereture\ Difference$

Used in



Results & Learning

- ✓ Learned practical engineering calculations
- ✓ Improved programing skills
- ✓ Concepted theory with application
- ✓ Developed problem-solving mindset

Conclusion

Integrating C programming streamlines complex chemical analysis and drives industrial efficiency.

Future Scope:

while this project currently uses a terminal-based C interface, the future scope involves building a full GUI and integrating Python for advanced data visualization and AI-driven process optimization.

This project demonstrates that C programming isn't just for software – it's a vital tool for modern chemical engineers to ensure accuracy and safety in industrial processes.